Supplementary material

Uncertainty-aware space–ground sensor fusion for tracking maneuvering hypersonic targets: A reproducible simulation benchmark

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S1. Contents

This supplement extends the results reported in the paper. Section and table numbers without the prefix S refer to the paper.

- Tables S1–S3: every sensing configuration and estimator at each maneuver level, with all metrics and confidence intervals (Section 7.2 and Table 5 report the medium level).
- Table S4: all 36 paired estimator comparisons; Table S5: all 24 paired comparisons of fused against single-sensor tracking (Sections 7.2–7.3; Table 6 reports the fused subset).
- Tables S6–S7: credibility diagnostics of the stated covariance for every estimator (Section 7.4).
- Tables S8–S9: every value of the sensitivity sweeps for all four estimators (Section 7.5 and Fig. 9).
- Table S10: the complete process-noise tuning grid (Appendix A and Table A.1); Table S11: statistics of the held-out scenarios (Section 4); Table S12: computational cost of every cell (Section 7.6).
- Figure S1: mean OSPA and track availability over time; Figure S2: distributions of the paired per-run OSPA differences behind Table 6.
- Section S7: implementation choices where the paper leaves a detail open; Table S13: the 31 automated verification tests; Section S9: how to reproduce every number.

Every value in this supplement is generated by `make_supplement.py` from the released result files; none was entered by hand. The same tables are provided as CSV files (Supplementary data, `Supplementary_Data.zip`, and `results/supplement/` in the repository). The code, configurations, seeds and results are openly available at `https://github.com/labibibnmuzahid/hypershield-benchmark`.

S2. Results for every configuration and estimator

Tables S1–S3 list all 36 cells of the main evaluation (3 sensing configurations $\times$ 4 estimators $\times$ 3 maneuver levels, 500 held-out runs per maneuver level). Figure S1 shows how the advantage of fusion builds up over the 480 s scenario: the IR layer provides a track within seconds, the radar contributes only after the target enters its coverage, and the fused tracker keeps the lowest OSPA throughout.

Table S1. All sensing configurations and estimators at the low (0.5 g) maneuver level (500 held-out runs each).

Estimator	OSPA (km) [95% CI]	Custody (%)	Contin. (%)	Segments per run	TTFT (s)	Ever tracked (%)	RMSE (km)	ANEES [95% CI]	C_{50}	C_{95}
IR only										
EKF-CV	2.13 [2.02, 2.23]	77.4	83.6	3.84	4	100.0	3.59	2.86 [2.66, 3.08]	0.578	0.924
UKF-CV	2.13 [2.02, 2.23]	77.3	83.5	3.85	4	100.0	3.60	2.86 [2.66, 3.08]	0.577	0.924
EKF-CA	2.27 [2.15, 2.37]	74.6	83.3	4.29	4	100.0	4.40	3.03 [2.83, 3.23]	0.522	0.899
IMM-CV/CA	2.05 [1.94, 2.16]	78.6	83.9	3.76	4	100.0	3.49	2.40 [2.24, 2.57]	0.525	0.918
Radar only										
EKF-CV	6.37 [6.25, 6.49]	36.7	38.0	1.35	285	99.4	0.61	2.29 [1.96, 2.67]	0.558	0.958
UKF-CV	6.37 [6.25, 6.49]	36.7	38.0	1.36	285	99.4	0.61	2.17 [1.87, 2.51]	0.558	0.958
EKF-CA	6.38 [6.26, 6.50]	36.1	38.1	1.48	283.5	99.4	0.66	2.37 [2.09, 2.69]	0.533	0.951
IMM-CV/CA	6.35 [6.23, 6.48]	36.9	38.1	1.35	284	99.4	0.59	2.27 [1.99, 2.59]	0.516	0.946
Fused										
EKF-CV	1.46 [1.38, 1.55]	85.1	89.3	3.05	4	100.0	2.94	2.31 [2.15, 2.48]	0.570	0.938
UKF-CV	1.46 [1.38, 1.55]	85.1	89.3	3.06	4	100.0	2.94	2.29 [2.14, 2.46]	0.570	0.938
EKF-CA	1.56 [1.47, 1.64]	83.2	89.1	3.37	4	100.0	3.66	2.39 [2.24, 2.55]	0.525	0.919
IMM-CV/CA	1.40 [1.32, 1.49]	85.9	89.6	3.00	4	100.0	2.88	2.01 [1.89, 2.14]	0.520	0.929

OSPA: mean OSPA distance, cut-off 10 km. Custody: share of epochs with a confirmed track within 1 km. Contin.: share of epochs with a confirmed track. Segments: confirmed track segments per run. TTFT: median time to first track within 5 km (runs never tracked count as infinite; a median over an even number of runs can end in .5, which the paper rounds to whole seconds). Ever tracked: share of runs with at least one such track. RMSE: over epochs with a track. ANEES: normalized (ideal 1). C_α: empirical coverage of the nominal α ellipsoid. Confidence intervals: run-level percentile bootstrap, 2000 resamples.

Table S2. All sensing configurations and estimators at the medium (2 g) maneuver level (500 held-out runs each).

Estimator	OSPA (km) [95% CI]	Custody (%)	Contin. (%)	Segments per run	TTFT (s)	Ever tracked (%)	RMSE (km)	ANEES [95% CI]	C_{50}	C_{95}
IR only										
EKF-CV	2.21 [2.10, 2.32]	75.5	83.2	3.94	4	100.0	4.02	3.18 [2.94, 3.43]	0.547	0.916
UKF-CV	2.20 [2.09, 2.32]	75.5	83.2	3.94	4	100.0	4.02	3.15 [2.92, 3.39]	0.547	0.916
EKF-CA	2.32 [2.21, 2.44]	73.6	83.0	4.37	4	100.0	4.74	3.22 [3.01, 3.46]	0.508	0.891
IMM-CV/CA	2.13 [2.02, 2.24]	76.7	83.6	3.81	4	100.0	4.00	2.61 [2.45, 2.77]	0.476	0.894
Radar only										
EKF-CV	6.48 [6.36, 6.60]	35.3	36.9	1.40	293.5	98.8	0.66	2.66 [2.30, 3.09]	0.532	0.950
UKF-CV	6.48 [6.37, 6.60]	35.3	36.9	1.39	293.5	98.8	0.66	2.48 [2.15, 2.87]	0.532	0.950
EKF-CA	6.48 [6.37, 6.60]	34.8	37.0	1.50	293	98.8	0.70	2.45 [2.15, 2.78]	0.524	0.945
IMM-CV/CA	6.48 [6.36, 6.59]	35.3	37.0	1.40	293.5	98.8	0.66	2.48 [2.17, 2.84]	0.474	0.927
Fused										
EKF-CV	1.56 [1.48, 1.65]	83.4	88.6	3.15	4	100.0	3.19	2.58 [2.39, 2.79]	0.546	0.930
UKF-CV	1.56 [1.48, 1.64]	83.5	88.6	3.15	4	100.0	3.18	2.56 [2.36, 2.76]	0.546	0.931
EKF-CA	1.64 [1.56, 1.73]	82.2	88.4	3.49	4	100.0	3.70	2.64 [2.46, 2.83]	0.515	0.913
IMM-CV/CA	1.51 [1.43, 1.59]	84.1	88.9	3.07	4	100.0	3.23	2.30 [2.16, 2.46]	0.477	0.905

OSPA: mean OSPA distance, cut-off 10 km. Custody: share of epochs with a confirmed track within 1 km. Contin.: share of epochs with a confirmed track. Segments: confirmed track segments per run. TTFT: median time to first track within 5 km (runs never tracked count as infinite; a median over an even number of runs can end in .5, which the paper rounds to whole seconds). Ever tracked: share of runs with at least one such track. RMSE: over epochs with a track. ANEES: normalized (ideal 1). C_α: empirical coverage of the nominal α ellipsoid. Confidence intervals: run-level percentile bootstrap, 2000 resamples.

Table S3. All sensing configurations and estimators at the high (4 g) maneuver level (500 held-out runs each).

Estimator	OSPA (km) [95% CI]	Custody (%)	Contin. (%)	Segments per run	TTFT (s)	Ever tracked (%)	RMSE (km)	ANEES [95% CI]	C_{50}	C_{95}
IR only										
EKF-CV	2.29 [2.18, 2.40]	73.8	83.6	4.13	4	100.0	4.14	3.81 [3.54, 4.07]	0.468	0.871
UKF-CV	2.29 [2.17, 2.40]	73.8	83.6	4.13	4	100.0	4.12	3.79 [3.53, 4.06]	0.468	0.871
EKF-CA	2.33 [2.22, 2.44]	73.1	83.5	4.42	4	100.0	4.75	3.62 [3.41, 3.86]	0.466	0.861
IMM-CV/CA	2.23 [2.12, 2.34]	74.6	83.8	4.07	4	100.0	4.14	3.40 [3.19, 3.62]	0.430	0.848
Radar only										
EKF-CV	6.69 [6.56, 6.83]	32.4	34.9	1.37	300.5	96.4	0.69	2.41 [2.06, 2.77]	0.469	0.921
UKF-CV	6.69 [6.56, 6.83]	32.5	34.9	1.37	300.5	96.4	0.69	2.25 [1.93, 2.59]	0.470	0.921
EKF-CA	6.68 [6.55, 6.82]	32.5	35.0	1.46	300	96.6	0.72	2.24 [1.97, 2.52]	0.494	0.935
IMM-CV/CA	6.68 [6.55, 6.82]	32.7	34.9	1.38	300.5	96.4	0.68	2.26 [1.98, 2.56]	0.451	0.910
Fused										
EKF-CV	1.62 [1.53, 1.70]	82.1	88.9	3.36	4	100.0	3.29	3.05 [2.83, 3.27]	0.481	0.895
UKF-CV	1.62 [1.53, 1.70]	82.2	88.9	3.36	4	100.0	3.28	3.02 [2.81, 3.24]	0.481	0.896
EKF-CA	1.65 [1.56, 1.73]	81.6	88.9	3.62	4	100.0	3.80	2.92 [2.74, 3.12]	0.476	0.886
IMM-CV/CA	1.57 [1.49, 1.66]	82.7	89.1	3.34	4	100.0	3.33	2.84 [2.67, 3.04]	0.440	0.872

OSPA: mean OSPA distance, cut-off 10 km. Custody: share of epochs with a confirmed track within 1 km. Contin.: share of epochs with a confirmed track. Segments: confirmed track segments per run. TTFT: median time to first track within 5 km (runs never tracked count as infinite; a median over an even number of runs can end in .5, which the paper rounds to whole seconds). Ever tracked: share of runs with at least one such track. RMSE: over epochs with a track. ANEES: normalized (ideal 1). C_α: empirical coverage of the nominal α ellipsoid. Confidence intervals: run-level percentile bootstrap, 2000 resamples.

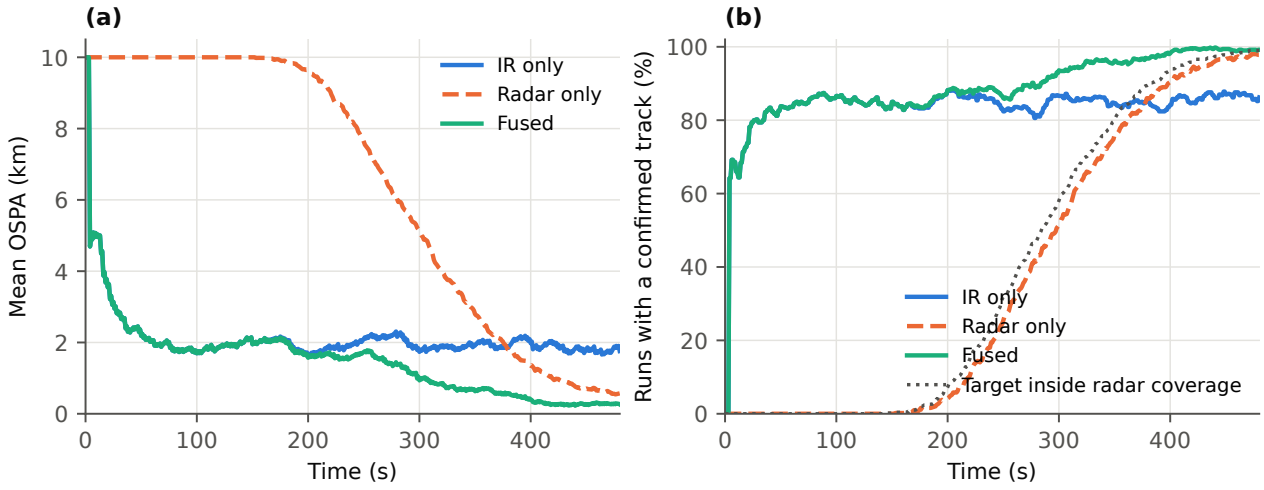


Figure S1. Accuracy and track availability over time, IMM-CV/CA at the medium maneuver level (500 runs). (a) Mean OSPA at each epoch (cut-off 10 km). (b) Share of runs holding a confirmed track; the dotted line is the share of runs whose target is inside radar coverage.

S3. Paired comparisons

Table S4 lists every estimator comparison in the Holm family used in the paper, and Table S5 every comparison of fused against single-sensor tracking. Figure S2 shows the run-level distributions behind the "Better" column of Table 6 for the IMM against EKF-CV: most differences are small and negative, which is why a 3–4% mean improvement is highly significant in the paired test.

Table S4. All 36 paired estimator comparisons of per-run OSPA (one Holm family).

Maneuver	Sensing	Comparison	Δ OSPA (km)	Δ (%)	Better (%)	p	Holm p
Low	IR only	IMM-CV/CA vs EKF-CV	−0.0794	−3.73	80.6	1.8×10^{-32}	5.0×10^{-31}
Low	IR only	IMM-CV/CA vs UKF-CV	−0.0821	−3.85	81.0	1.1×10^{-33}	3.3×10^{-32}
Low	IR only	IMM-CV/CA vs EKF-CA	−0.2162	−9.54	86.6	2.0×10^{-44}	6.8×10^{-43}
Low	IR only	UKF-CV vs EKF-CV	+0.0028	+0.13	54.2	0.007	0.037
Low	Radar only	IMM-CV/CA vs EKF-CV	−0.0160	−0.25	71.8	1.5×10^{-29}	3.9×10^{-28}
Low	Radar only	IMM-CV/CA vs UKF-CV	−0.0162	−0.25	71.6	4.5×10^{-29}	1.2×10^{-27}
Low	Radar only	IMM-CV/CA vs EKF-CA	−0.0241	−0.38	86.2	1.2×10^{-46}	4.2×10^{-45}
Low	Radar only	UKF-CV vs EKF-CV	+0.0002	+0.00	36.6	5.8×10^{-11}	9.2×10^{-10}
Low	Fused	IMM-CV/CA vs EKF-CV	−0.0568	−3.89	83.0	5.6×10^{-33}	1.7×10^{-31}
Low	Fused	IMM-CV/CA vs UKF-CV	−0.0560	−3.83	83.4	6.9×10^{-34}	2.2×10^{-32}
Low	Fused	IMM-CV/CA vs EKF-CA	−0.1508	−9.70	85.8	1.7×10^{-40}	5.9×10^{-39}
Low	Fused	UKF-CV vs EKF-CV	−0.0008	−0.05	52.4	0.008	0.037
Medium	IR only	IMM-CV/CA vs EKF-CV	−0.0795	−3.60	77.6	4.4×10^{-25}	1.1×10^{-23}
Medium	IR only	IMM-CV/CA vs UKF-CV	−0.0770	−3.49	77.4	2.1×10^{-24}	5.0×10^{-23}
Medium	IR only	IMM-CV/CA vs EKF-CA	−0.1971	−8.48	79.4	3.4×10^{-35}	1.1×10^{-33}
Medium	IR only	UKF-CV vs EKF-CV	−0.0024	−0.11	56.4	1.2×10^{-5}	9.7×10^{-5}
Medium	Radar only	IMM-CV/CA vs EKF-CV	−0.0034	−0.05	53.2	0.016	0.047
Medium	Radar only	IMM-CV/CA vs UKF-CV	−0.0054	−0.08	53.2	0.020	0.047
Medium	Radar only	IMM-CV/CA vs EKF-CA	−0.0081	−0.12	72.6	2.7×10^{-21}	5.6×10^{-20}
Medium	Radar only	UKF-CV vs EKF-CV	+0.0021	+0.03	33.8	4.2×10^{-15}	7.1×10^{-14}
Medium	Fused	IMM-CV/CA vs EKF-CV	−0.0537	−3.43	76.0	5.6×10^{-23}	1.3×10^{-21}
Medium	Fused	IMM-CV/CA vs UKF-CV	−0.0510	−3.26	75.8	1.4×10^{-22}	3.2×10^{-21}
Medium	Fused	IMM-CV/CA vs EKF-CA	−0.1316	−8.01	78.6	9.7×10^{-33}	2.8×10^{-31}
Medium	Fused	UKF-CV vs EKF-CV	−0.0027	−0.17	56.2	2.9×10^{-5}	2.1×10^{-4}
High	IR only	IMM-CV/CA vs EKF-CV	−0.0602	−2.63	68.4	3.9×10^{-9}	4.7×10^{-8}
High	IR only	IMM-CV/CA vs UKF-CV	−0.0554	−2.42	67.8	2.6×10^{-8}	2.9×10^{-7}
High	IR only	IMM-CV/CA vs EKF-CA	−0.1022	−4.38	65.0	4.1×10^{-10}	5.8×10^{-9}
High	IR only	UKF-CV vs EKF-CV	−0.0048	−0.21	54.2	0.004	0.021
High	Radar only	IMM-CV/CA vs EKF-CV	−0.0122	−0.18	65.8	3.4×10^{-20}	6.8×10^{-19}
High	Radar only	IMM-CV/CA vs UKF-CV	−0.0086	−0.13	65.6	2.7×10^{-19}	5.1×10^{-18}
High	Radar only	IMM-CV/CA vs EKF-CA	+0.0027	+0.04	61.2	7.1×10^{-6}	6.4×10^{-5}
High	Radar only	UKF-CV vs EKF-CV	−0.0036	−0.05	33.0	2.2×10^{-15}	4.0×10^{-14}
High	Fused	IMM-CV/CA vs EKF-CV	−0.0471	−2.90	69.6	1.2×10^{-10}	1.7×10^{-9}
High	Fused	IMM-CV/CA vs UKF-CV	−0.0430	−2.66	69.2	1.2×10^{-9}	1.5×10^{-8}
High	Fused	IMM-CV/CA vs EKF-CA	−0.0745	−4.52	62.2	6.5×10^{-8}	6.5×10^{-7}
High	Fused	UKF-CV vs EKF-CV	−0.0041	−0.25	54.6	0.061	0.061

Δ: mean per-run OSPA difference, first minus second estimator; Δ (%): relative to the second estimator. Better: share of runs in which the first estimator had the lower OSPA. p: two-sided Wilcoxon signed-rank test on 500 paired runs; Holm p: adjusted over all 36 comparisons.

Table S5. All 24 paired comparisons of fused against single-sensor tracking (per-run OSPA, one Holm family).

Maneuver	Estimator	Comparison	Δ OSPA (km)	Δ (%)	Better (%)	p	Holm p
Low	EKF-CV	Fused vs IR only	−0.667	−31.3	97.8	1.2×10^{-79}	1.3×10^{-78}
Low	EKF-CV	Fused vs Radar only	−4.905	−77.0	100.0	1.3×10^{-83}	3.0×10^{-82}
Low	UKF-CV	Fused vs IR only	−0.670	−31.4	97.8	1.2×10^{-79}	1.3×10^{-78}
Low	UKF-CV	Fused vs Radar only	−4.906	−77.1	100.0	1.3×10^{-83}	3.0×10^{-82}
Low	EKF-CA	Fused vs IR only	−0.709	−31.3	98.0	1.1×10^{-80}	1.3×10^{-79}
Low	EKF-CA	Fused vs Radar only	−4.819	−75.6	100.0	1.3×10^{-83}	3.0×10^{-82}
Low	IMM-CV/CA	Fused vs IR only	−0.644	−31.4	97.4	5.3×10^{-79}	4.7×10^{-78}
Low	IMM-CV/CA	Fused vs Radar only	−4.946	−77.9	100.0	1.3×10^{-83}	3.0×10^{-82}
Medium	EKF-CV	Fused vs IR only	−0.642	−29.1	97.0	2.0×10^{-78}	1.6×10^{-77}
Medium	EKF-CV	Fused vs Radar only	−4.915	−75.8	100.0	1.3×10^{-83}	3.0×10^{-82}
Medium	UKF-CV	Fused vs IR only	−0.642	−29.1	97.0	2.2×10^{-78}	1.6×10^{-77}
Medium	UKF-CV	Fused vs Radar only	−4.920	−75.9	100.0	1.3×10^{-83}	3.0×10^{-82}
Medium	EKF-CA	Fused vs IR only	−0.681	−29.3	97.0	2.3×10^{-78}	1.6×10^{-77}
Medium	EKF-CA	Fused vs Radar only	−4.842	−74.7	100.0	1.3×10^{-83}	3.0×10^{-82}
Medium	IMM-CV/CA	Fused vs IR only	−0.616	−29.0	96.8	3.0×10^{-77}	6.1×10^{-77}
Medium	IMM-CV/CA	Fused vs Radar only	−4.965	−76.7	100.0	1.3×10^{-83}	3.0×10^{-82}
High	EKF-CV	Fused vs IR only	−0.668	−29.2	95.6	5.7×10^{-78}	2.3×10^{-77}
High	EKF-CV	Fused vs Radar only	−5.072	−75.8	100.0	1.3×10^{-83}	3.0×10^{-82}
High	UKF-CV	Fused vs IR only	−0.668	−29.2	95.6	5.7×10^{-78}	2.3×10^{-77}
High	UKF-CV	Fused vs Radar only	−5.073	−75.8	100.0	1.3×10^{-83}	3.0×10^{-82}
High	EKF-CA	Fused vs IR only	−0.683	−29.3	95.2	2.0×10^{-78}	1.6×10^{-77}
High	EKF-CA	Fused vs Radar only	−5.030	−75.3	100.0	1.3×10^{-83}	3.0×10^{-82}
High	IMM-CV/CA	Fused vs IR only	−0.655	−29.4	94.8	8.3×10^{-77}	8.3×10^{-77}
High	IMM-CV/CA	Fused vs Radar only	−5.107	−76.4	100.0	1.3×10^{-83}	3.0×10^{-82}

Δ: mean per-run OSPA difference, fused minus single-sensor; Δ (%): relative to the single-sensor value. Better: share of runs in which fused tracking had the lower OSPA. p: two-sided Wilcoxon signed-rank test; Holm p: adjusted over the 24 comparisons.

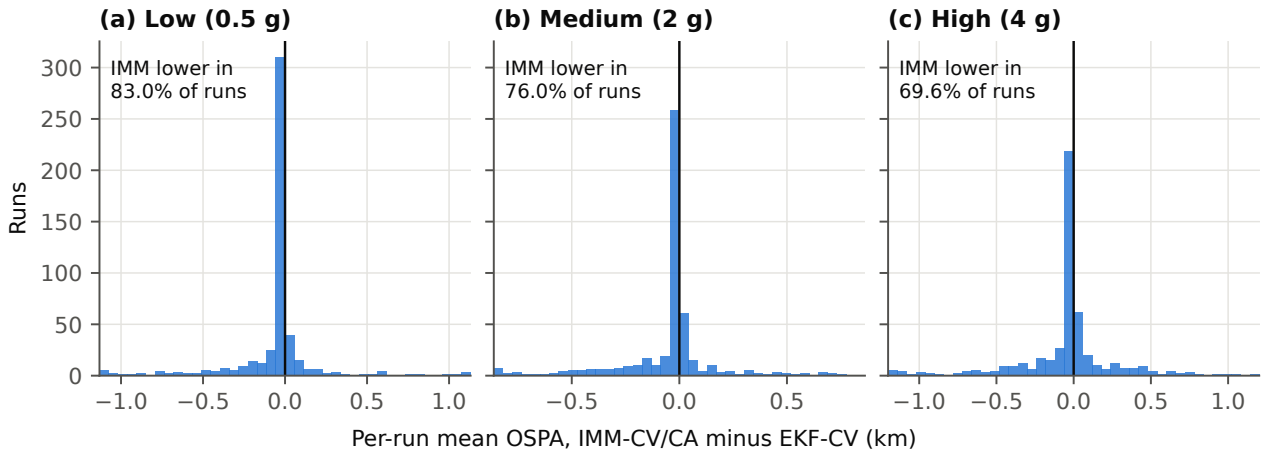


Figure S2. Distribution of the per-run OSPA difference between IMM-CV/CA and EKF-CV in the fused configuration (500 paired runs per maneuver level). Negative values favor the IMM. Differences beyond the 99th percentile of their absolute value are drawn at the axis limits.

S4. Credibility diagnostics

Table S6 extends Fig. 8 of the paper to all four estimators, and Table S7 gives the acceleration-binned ANEES quoted in Section 7.4.

Table S6. Credibility diagnostics of the stated position covariance, fused configuration, medium maneuver level.

Quantity	EKF-CV	UKF-CV	EKF-CA	IMM-CV/CA
Epochs within 10 s after a false association (%)	18.1	18.1	18.2	16.6
Normalized ANEES after a false association	9.42	9.28	9.55	8.22
Normalized ANEES, all other epochs	1.07	1.07	1.10	1.12
C_{95} after a false association	0.773	0.774	0.728	0.749
C_{95}, all other epochs	0.965	0.965	0.954	0.936
Share of summed NEES after a false association (%)	66.1	65.8	65.9	59.5
Share of updates that used a clutter detection (%)	1.43	1.43	1.52	1.33
Median normalized NEES	0.71	0.71	0.76	0.83
Share of summed NEES in the worst 5% of epochs (%)	69.3	69.0	67.0	59.4
C_{50} (nominal 0.50)	0.546	0.546	0.515	0.477
C_{68} (nominal 0.68)	0.713	0.713	0.682	0.644
C_{90} (nominal 0.90)	0.894	0.894	0.872	0.856
C_{95} (nominal 0.95)	0.930	0.931	0.913	0.905
C_{99} (nominal 0.99)	0.958	0.958	0.947	0.952

A false association is an update (or a track initiation) that used a clutter detection. Normalized NEES is NEES/3, so its consistent mean is 1. C_α: empirical coverage of the nominal α ellipsoid over all scored epochs.

Table S7. Normalized ANEES by the target's instantaneous lateral acceleration, fused configuration, medium maneuver level.

Lateral acceleration	Epochs (%)	EKF-CV	UKF-CV	EKF-CA	IMM-CV/CA
0–0.5 g (0.0–4.9 m/s^2)	38.6	2.38	2.35	2.50	2.08
0.5–1 g (4.9–9.8 m/s^2)	32.4	2.64	2.61	2.68	2.32
1–1.5 g (9.8–14.7 m/s^2)	20.4	3.03	3.01	3.00	2.67
$>$1.5 g ($>$14.7 m/s^2)	8.6	2.21	2.20	2.21	2.32

Epochs (%): share of scored epochs (IMM-CV/CA) in each bin. Compare with Table S6: the spread across acceleration bins is far smaller than the difference between epochs with and without a recent false association.

S5. Sensitivity analysis

Tables S8 and S9 give every value plotted in Fig. 9 of the paper, for all four estimators.

Table S8. One-at-a-time sensitivity analysis, fused configuration (300 medium-maneuver runs with trajectories fixed across parameter values).

	Mean OSPA (km)				Normalized ANEES				False
Value	EKF-CV	UKF-CV	EKF-CA	IMM	EKF-CV	UKF-CV	EKF-CA	IMM	upd. (%)
IR revisit interval (s)									
1	1.04	1.04	1.07	1.02	1.67	1.66	1.75	1.69	0.97
2*	1.39	1.39	1.49	1.34	2.25	2.22	2.47	2.07	1.29
5	2.34	2.32	2.93	2.20	3.82	3.67	6.57	3.49	2.30
10	6.43	6.43	6.45	6.43	2.33	2.21	2.24	2.28	1.28
IR outage fraction (%)									
0	0.49	0.49	0.54	0.45	1.55	1.53	1.80	1.54	1.08
10	0.96	0.96	1.04	0.92	1.83	1.80	2.09	1.77	1.18
20*	1.39	1.39	1.49	1.34	2.25	2.22	2.47	2.07	1.29
35	2.20	2.20	2.33	2.15	3.06	3.01	3.25	2.75	1.54
50	3.06	3.05	3.18	3.00	4.00	3.89	3.98	3.35	1.85
IR angle noise (μrad)									
50	1.27	1.28	1.29	1.24	3.12	3.12	2.86	2.69	0.66
100*	1.39	1.39	1.49	1.34	2.25	2.22	2.47	2.07	1.29
150	1.46	1.45	1.58	1.43	2.10	2.08	2.40	2.04	2.20
200	1.49	1.49	1.64	1.44	1.93	1.92	2.30	1.90	3.35
Radar detection probability									
0.5	2.13	2.13	2.27	2.08	6.55	6.30	6.49	5.88	2.76
0.6	1.78	1.78	1.90	1.71	4.21	4.19	4.25	3.61	2.07
0.7	1.55	1.54	1.66	1.49	3.08	2.99	3.14	2.71	1.66
0.8	1.44	1.43	1.54	1.38	2.43	2.36	2.72	2.21	1.43
0.9*	1.39	1.39	1.49	1.34	2.25	2.22	2.47	2.07	1.29
False alarms per scan									
0	0.98	0.98	1.00	0.97	0.84	0.84	0.89	1.01	0.00
0.5	1.15	1.15	1.19	1.13	1.41	1.40	1.40	1.42	0.57
1*	1.39	1.39	1.49	1.34	2.25	2.22	2.47	2.07	1.29
2	2.16	2.13	2.29	1.97	6.83	6.68	6.92	5.09	3.21
3	2.96	2.95	3.37	2.80	15.19	14.67	16.18	11.75	6.20

* Nominal value (Table 3 of the paper). IMM: IMM-CV/CA. False upd.: share of IMM-CV/CA updates that used a clutter detection.

Table S9. One-at-a-time sensitivity analysis, IR-only configuration (300 medium-maneuver runs with trajectories fixed across parameter values).

	Mean OSPA (km)				Normalized ANEES				False
Value	EKF-CV	UKF-CV	EKF-CA	IMM	EKF-CV	UKF-CV	EKF-CA	IMM	upd. (%)
IR revisit interval (s)									
1	1.46	1.46	1.49	1.44	1.88	1.88	1.89	1.90	1.12
2*	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
5	3.28	3.25	4.11	3.07	5.30	5.07	9.55	4.69	4.80
10	10.00	10.00	10.00	10.00	–	–	–	–	0.00
IR outage fraction (%)									
0	0.52	0.52	0.57	0.48	1.57	1.55	1.80	1.56	1.29
10	1.28	1.27	1.36	1.23	2.07	2.05	2.32	1.94	1.52
20*	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
35	3.19	3.19	3.39	3.13	3.90	3.89	4.25	3.33	2.33
50	4.53	4.53	4.75	4.47	5.95	5.86	5.75	4.78	3.29
IR angle noise (μrad)									
50	1.79	1.79	1.82	1.74	3.89	3.89	3.55	3.22	0.86
100*	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
150	2.03	2.03	2.21	1.98	2.47	2.46	2.88	2.29	3.10
200	2.06	2.06	2.28	1.99	2.24	2.24	2.72	2.11	4.71
Radar detection probability									
0.5	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
0.6	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
0.7	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
0.8	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
0.9*	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
False alarms per scan									
0	1.41	1.41	1.43	1.40	0.83	0.83	0.88	1.00	0.00
0.5	1.63	1.63	1.69	1.60	1.55	1.54	1.55	1.51	0.77
1*	1.92	1.92	2.05	1.85	2.73	2.70	2.96	2.38	1.76
2	2.86	2.84	3.12	2.67	8.03	7.87	8.38	6.14	4.67
3	3.92	3.90	4.45	3.75	16.70	16.46	18.38	13.68	9.32

* Nominal value (Table 3 of the paper). IMM: IMM-CV/CA. False upd.: share of IMM-CV/CA updates that used a clutter detection. Radar detection probability does not affect this configuration. At 10 s revisit no IR-only track is confirmed, so ANEES is undefined.

S6. Tuning, scenario statistics and computational cost

Table S10. Complete process-noise tuning grid (fused configuration, medium maneuvers, 200 tuning runs disjoint from all evaluation runs).

Estimator	Setting	Mean OSPA (km)	Custody (%)	ANEES	C_{95}	
EKF-CV	$q_v = 10^{-5}$	3.18	56.7	20.93	0.343	
EKF-CV	$q_v = 3\times10^{-5}$	2.59	65.1	12.65	0.431	
EKF-CV	$q_v = 10^{-4}$	1.70	80.1	4.80	0.690	
EKF-CV	$q_v = 3\times10^{-4}$	1.48	84.2	2.93	0.884	
EKF-CV	$q_v = 10^{-3}$	**1.45**	**84.7**	**2.39**	**0.932**	selected
EKF-CV	$q_v = 3\times10^{-3}$	1.50	83.7	2.35	0.930	
EKF-CV	$q_v = 10^{-2}$	1.58	81.7	2.59	0.917	
EKF-CA	$q_a = 10^{-7}$	2.46	68.3	9.45	0.546	
EKF-CA	$q_a = 10^{-6}$	1.59	82.1	3.32	0.801	
EKF-CA	$q_a = 10^{-5}$	**1.52**	**83.5**	**2.39**	**0.914**	selected
EKF-CA	$q_a = 10^{-4}$	1.67	81.0	2.61	0.909	
EKF-CA	$q_a = 10^{-3}$	2.13	74.3	4.20	0.864	
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-5}$, $p_s = 0.90$	**1.41**	**85.2**	**2.22**	**0.905**	selected
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-5}$, $p_s = 0.95$	1.41	85.1	2.18	0.904	
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-5}$, $p_s = 0.98$	1.43	84.8	2.27	0.893	
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-4}$, $p_s = 0.90$	1.44	84.6	2.09	0.928	
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-4}$, $p_s = 0.95$	1.44	84.6	2.14	0.913	
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-4}$, $p_s = 0.98$	1.47	84.0	2.38	0.880	
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-3}$, $p_s = 0.90$	1.42	84.7	1.90	0.939	
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-3}$, $p_s = 0.95$	1.45	84.5	2.06	0.917	
IMM-CV/CA	$q_v = 10^{-5}$, $q_a = 10^{-3}$, $p_s = 0.98$	1.49	83.8	2.47	0.865	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-5}$, $p_s = 0.90$	1.41	85.0	2.23	0.912	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-5}$, $p_s = 0.95$	1.41	85.1	2.14	0.913	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-5}$, $p_s = 0.98$	1.42	84.8	2.24	0.906	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-4}$, $p_s = 0.90$	1.44	84.6	2.08	0.930	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-4}$, $p_s = 0.95$	1.44	84.7	2.09	0.923	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-4}$, $p_s = 0.98$	1.46	84.3	2.26	0.903	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-3}$, $p_s = 0.90$	1.42	84.6	1.89	0.940	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-3}$, $p_s = 0.95$	1.45	84.5	2.05	0.927	
IMM-CV/CA	$q_v = 10^{-4}$, $q_a = 10^{-3}$, $p_s = 0.98$	1.46	84.3	2.23	0.896	

q_v in km^2/s^3, q_a in km^2/s^5. The setting with the lowest mean OSPA was selected for each estimator; UKF-CV uses the EKF-CV value. Rows in bold are the selected settings.

Table S11. Statistics of the held-out scenarios (500 runs per maneuver level).

Quantity	Low (0.5 g)	Medium (2 g)	High (4 g)
Peak lateral acceleration, median (g)	0.49	1.95	3.91
Peak lateral acceleration, 5th–95th percentile (g)	0.38–0.61	1.54–2.39	3.06–4.93
Peak lateral acceleration, median (m/s^2)	4.8	19.1	38.4
Altitude range (km)	21.9–56.3	22.2–56.2	21.6–56.8
Mean number of visible satellites	5.25	5.23	5.26
Realized IR outage fraction (%)	20.7	21.2	20.3
Time inside radar coverage (%)	40.4	39.4	37.2
Runs never inside radar coverage (%)	0.6	1.0	3.0
Radar entry time, median (s)	276	284	289
Radar entry time, 5th–95th percentile (s)	184–400	193–402	187–429

$g = 9.80665$ m/s^2. Visible satellites: line of sight above the Earth's surface and range below 4500 km, averaged over IR scan times. Radar entry time: first epoch inside radar coverage, over the runs that enter it.

Table S12. Computational cost of each cell of the main evaluation.

		μs per filter cycle and run			Tracker CPU time (s)		
Sensing	Estimator	Low	Medium	High	Low	Medium	High
IR only	EKF-CV	5.7	6.7	5.1	1.37	1.62	1.23
IR only	UKF-CV	8.8	8.3	9.6	2.12	1.99	2.31
IR only	EKF-CA	7.4	7.6	7.4	1.77	1.82	1.77
IR only	IMM-CV/CA	16.0	16.7	14.9	3.84	4.01	3.58
Radar only	EKF-CV	2.3	1.7	2.2	0.55	0.40	0.52
Radar only	UKF-CV	3.4	3.0	3.0	0.81	0.71	0.72
Radar only	EKF-CA	2.8	2.8	2.3	0.66	0.68	0.56
Radar only	IMM-CV/CA	5.9	6.1	5.3	1.41	1.47	1.27
Fused	EKF-CV	8.0	7.7	6.3	1.92	1.85	1.51
Fused	UKF-CV	11.4	10.6	9.6	2.74	2.54	2.30
Fused	EKF-CA	9.5	9.3	8.0	2.29	2.23	1.93
Fused	IMM-CV/CA	20.2	18.8	18.4	4.86	4.51	4.42

Process CPU time of the tracker for 500 runs $\times$ 481 cycles, measured on one core with the vectorized Python implementation (scenario and detection generation excluded). Total over the 36 cells: 70.3 s. These figures compare estimators within one implementation and are not real-time performance claims.

S7. Implementation choices

Where the paper does not specify an implementation detail, the code makes the following choices. Each is visible in the released code.

1. **Heading-hold reference ψ_{ref}.** The heading of the initial great circle at the vehicle's current position.
2. **Altitude phase ϕ.** Drawn from $U[0, 2\pi)$.
3. **Tasking.** Uses the true target position, because detections are generated before any tracker runs.
4. **Outage chain.** Steps once per second. During an outage no IR satellite produces any detection, clutter included.
5. **Radar clutter.** Generated only while the target is inside radar coverage, centered on its true return.
6. **Initiation.** The initiating detection counts as the first of the three confirmation hits; the initiation cycle performs no further update.
7. **Association ratio.** Starts at 1 at initiation. The $\sqrt{\text{tr}(P_{\text{pos}})}$ deletion rule also applies to tentative tracks; the association-ratio rule applies to confirmed tracks only.
8. **False-association flag.** A track started from a clutter detection counts as a false association at its start.
9. **IMM details.** Initial mode probabilities are 0.5/0.5, and the acceleration block of the CV mode carries a variance of 10^{-12}.
10. **Estimator test family.** The 36 comparisons are {IMM vs EKF-CV, IMM vs UKF-CV, IMM vs EKF-CA, UKF-CV vs EKF-CV} $\times$ 3 sensing configurations $\times$ 3 maneuver levels.
11. **TTFT median.** Runs that are never tracked count as infinite.
12. **Sweep grids.** Revisit {1, 2, 5, 10} s; outage {0, 10, 20, 35, 50}%; angle noise {50, 100, 150, 200} μrad; radar detection probability {0.5, 0.6, 0.7, 0.8, 0.9}; false alarms {0, 0.5, 1, 2, 3} per scan, applied to both sensors.
13. **Example run (Figs. 2–3).** The first of 50 medium-maneuver runs from the example seed that enters radar coverage between 250 and 350 s with at least three IR outages before entry.

S8. Verification tests

The repository includes 31 automated test cases (28 test functions, two of them parametrized), run with `python -m pytest tests`. All pass with the software versions listed in Section S9.

Table S13. Automated verification tests.

Test	What it checks
Physics and geometry (`tests/test_physics_geometry.py`)	
`radar_line_of_sight_values`	Eq. (2) gives 596, 727 and 1023 km for targets at 20, 30 and 60 km (10 m mast)
`satellite_horizon`	Eq. (3) gives 3707 km for a satellite at 1000 km
`table1_values`	Stagnation temperatures, Wien peaks, plasma critical densities and warning times of Table 1
`destination_and_distance_roundtrip`	Great-circle destination and distance functions invert each other
`enu_basis_orthonormal`	The local east–north–up basis is orthonormal
`segment_clears_earth`	Earth occlusion of satellite–target lines of sight
`walker_constellation_shape`	The Walker 48/6/1 constellation has 48 distinct satellites at 1000 km altitude in orbits inclined at 55°
`ir_jacobian_matches_numeric`	Analytic IR angle Jacobian against finite differences
`radar_jacobian_and_inverse`	Analytic radar Jacobian against finite differences, and the polar-to-Cartesian inverse
`ir_los_unit_inverts_measurement`	Line-of-sight unit vectors recovered from the two IR angles
`mach_conversion`	Mach number to speed with a speed of sound of 0.303 km/s
Motion models and estimators (`tests/test_filters.py`)	
`process_noise_equals_integral` (3 cases)	CV and CA process-noise matrices equal the numerical integral of the continuous-time model for $T = 0.5$, 1 and 2 s
`ekf_and_ukf_exact_for_linear_measurement`	EKF and UKF updates reproduce the textbook Kalman filter for a linear measurement
`ukf_sigma_points_are_cubature_rule`	With $\alpha = 1$, $\beta = 0$, $\kappa = 0$ the unscented transform reproduces the exact second moments, as the cubature rule does
`filter_is_consistent_on_matched_model` (2 cases)	ANEES within 0.95–1.05 for EKF-CV and UKF-CV on a matched model
`imm_probabilities_and_consistency`	IMM mode probabilities sum to 1, covariances stay positive definite, the CV mode dominates on CV truth, and ANEES is within 0.8–1.2
`innovation_wraps_angles`	Angular innovations are wrapped across $\pm\pi$
Scenarios, sensors, tracker and statistics (`tests/test_pipeline.py`)	
`trajectory_envelope`	Initial speed within Mach 5–8, altitude bounds, position increments consistent with speed, and peak lateral acceleration bounded by the maneuver level
`maneuver_level_changes_only_lateral_command`	Changing the maneuver level changes only the lateral command
`outage_chain_statistics`	The Markov chain reproduces the 20% outage fraction and 25 s mean outage duration
`detection_statistics`	IR and radar detection probabilities of 0.9, one IR false alarm per scan, radar detections only inside coverage
`common_random_numbers_across_sweeps`	Changing one sensor parameter leaves the other sensor's data unchanged and nests the detections
`stereo_initiation_noise_free`	Noise-free stereo triangulation recovers the true position with a positive-definite covariance
`tracker_deterministic_and_consistent`	Repeated tracker runs are identical, and estimates exist only for confirmed tracks
`no_clutter_gives_credible_covariance`	Without clutter there are no false updates and the fused ANEES is below 1.5
`dataset_reproducible`	The same seed regenerates identical detections
`ospa_and_holm`	The single-target OSPA reduction and the Holm adjustment on a worked example
`bootstrap_interval_contains_point`	Bootstrap intervals contain the point estimate and have the expected width

S9. Reproducing the results

With Python 3.13, NumPy 2.5.3, SciPy 1.18.1 and Matplotlib 3.11.2 (`pip install -r requirements.txt`):

1. `python -m pytest -q tests` — the 31 verification tests;
2. `python run_experiments.py all` — tuning, main evaluation, sensitivity sweeps and the example run;
3. `python analysis_tables.py` and `python analysis_conditional.py` — the tables and every number quoted in the paper;
4. `python make_figs.py` — Figures 1–9;
5. `python make_supplement.py`, then `pdflatex Supplementary_Material.tex` in `supplement/` — this document.

Independent random streams come from NumPy `SeedSequence` instances with entropy 11 (tuning), 221, 222 and 223 (low, medium and high evaluation sets), 33 (sensitivity sweeps) and 44 (example run). A complete run regenerates every table, figure and number in the paper and in this supplement.